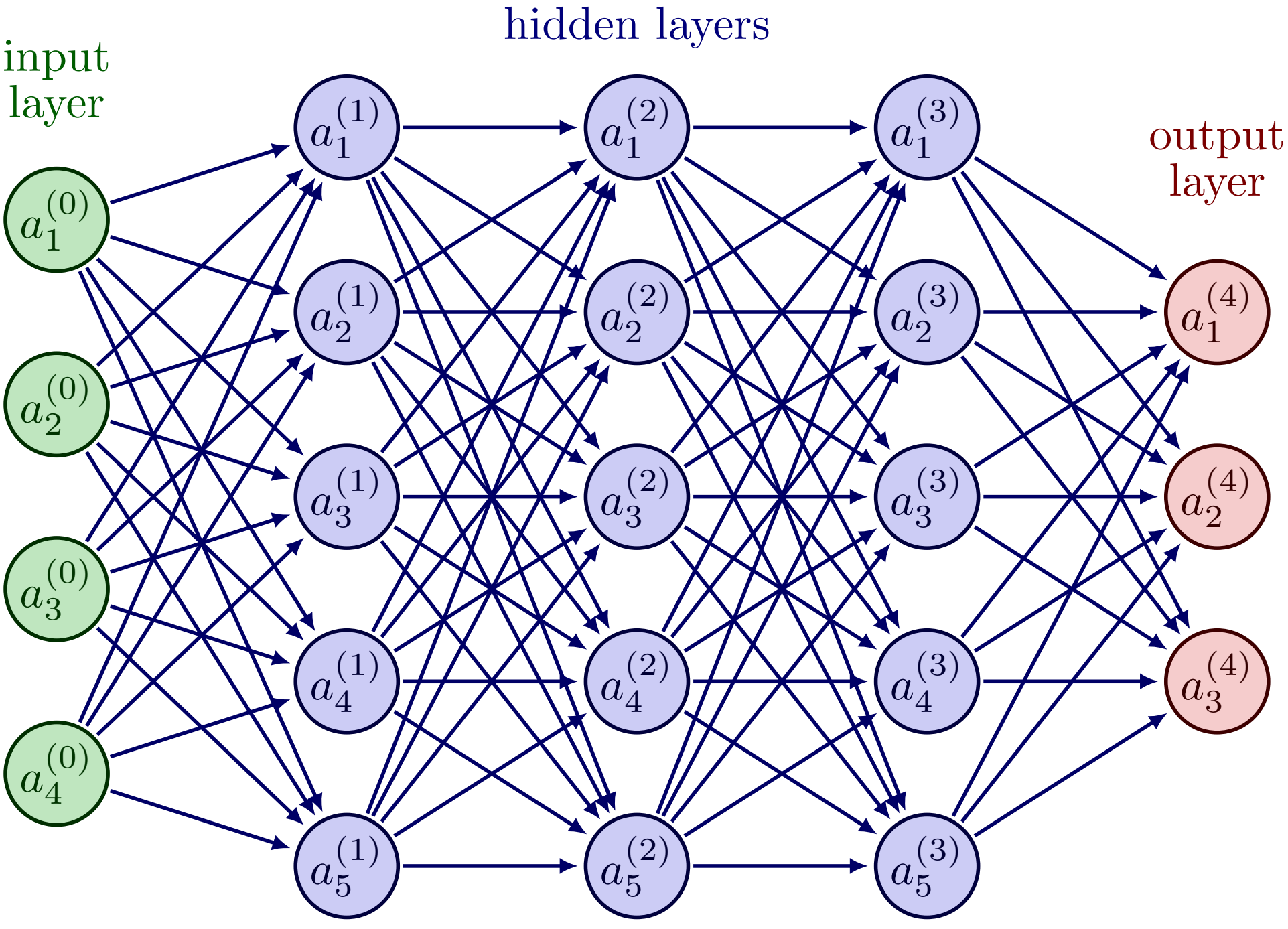
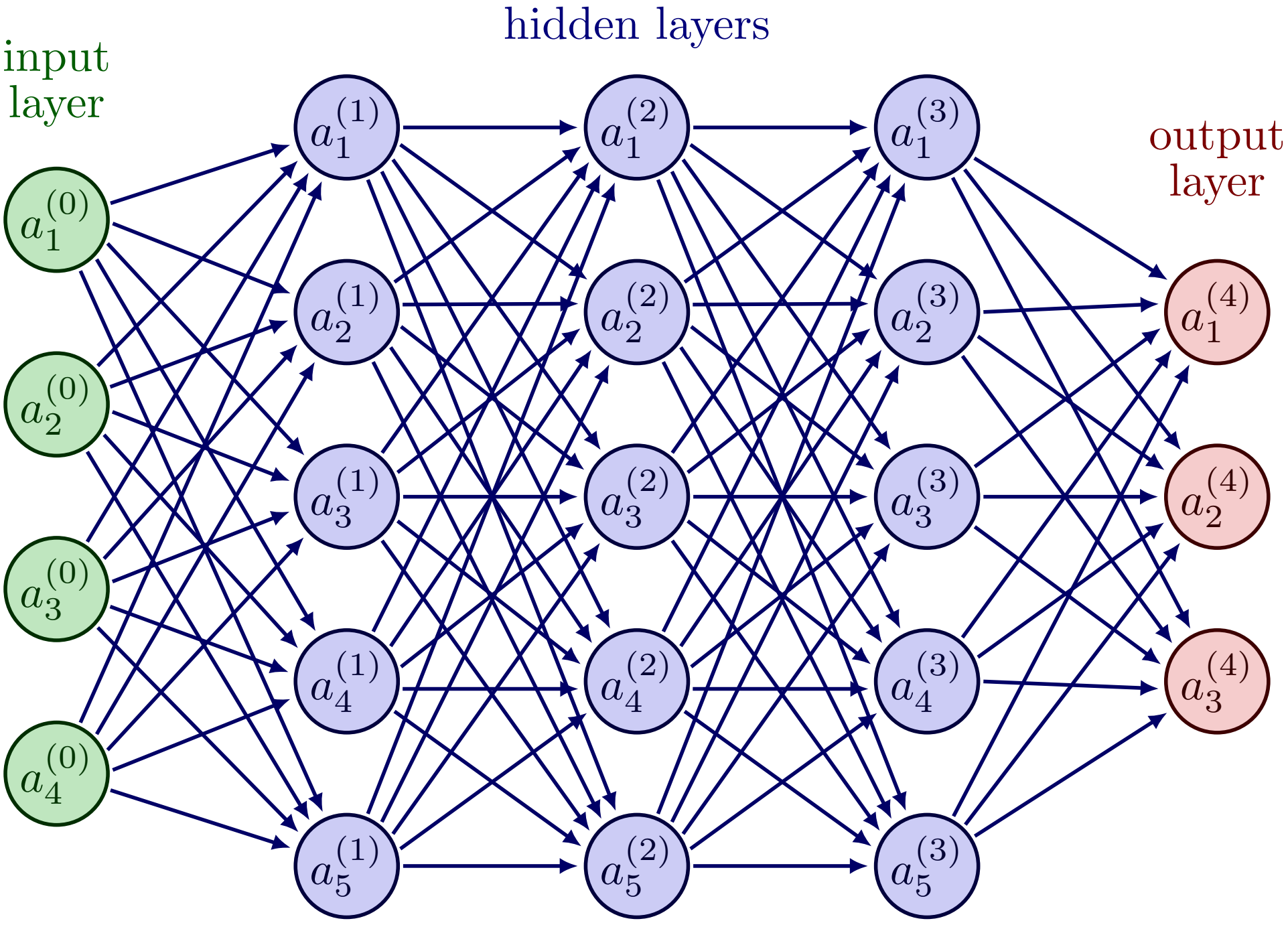
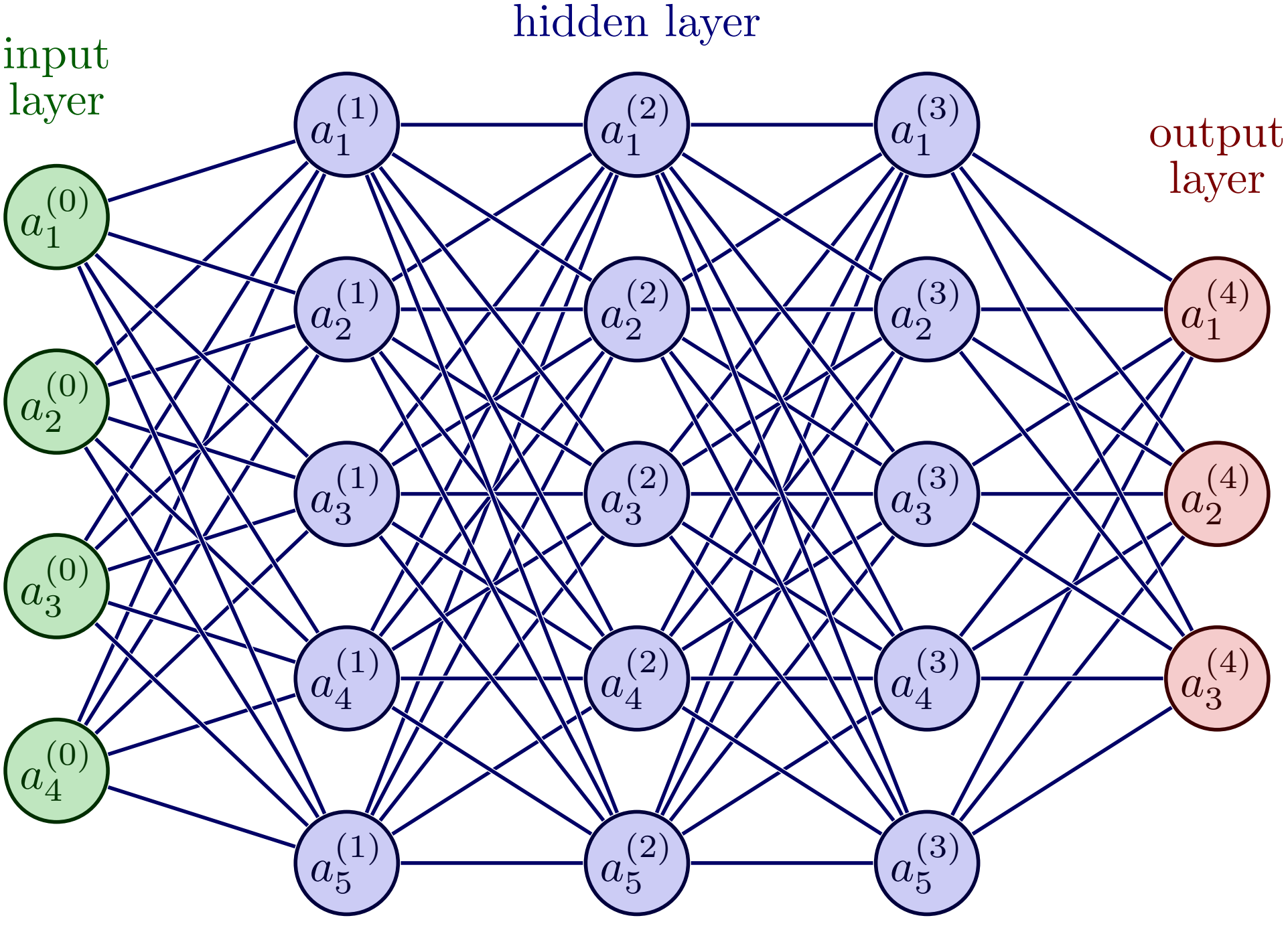
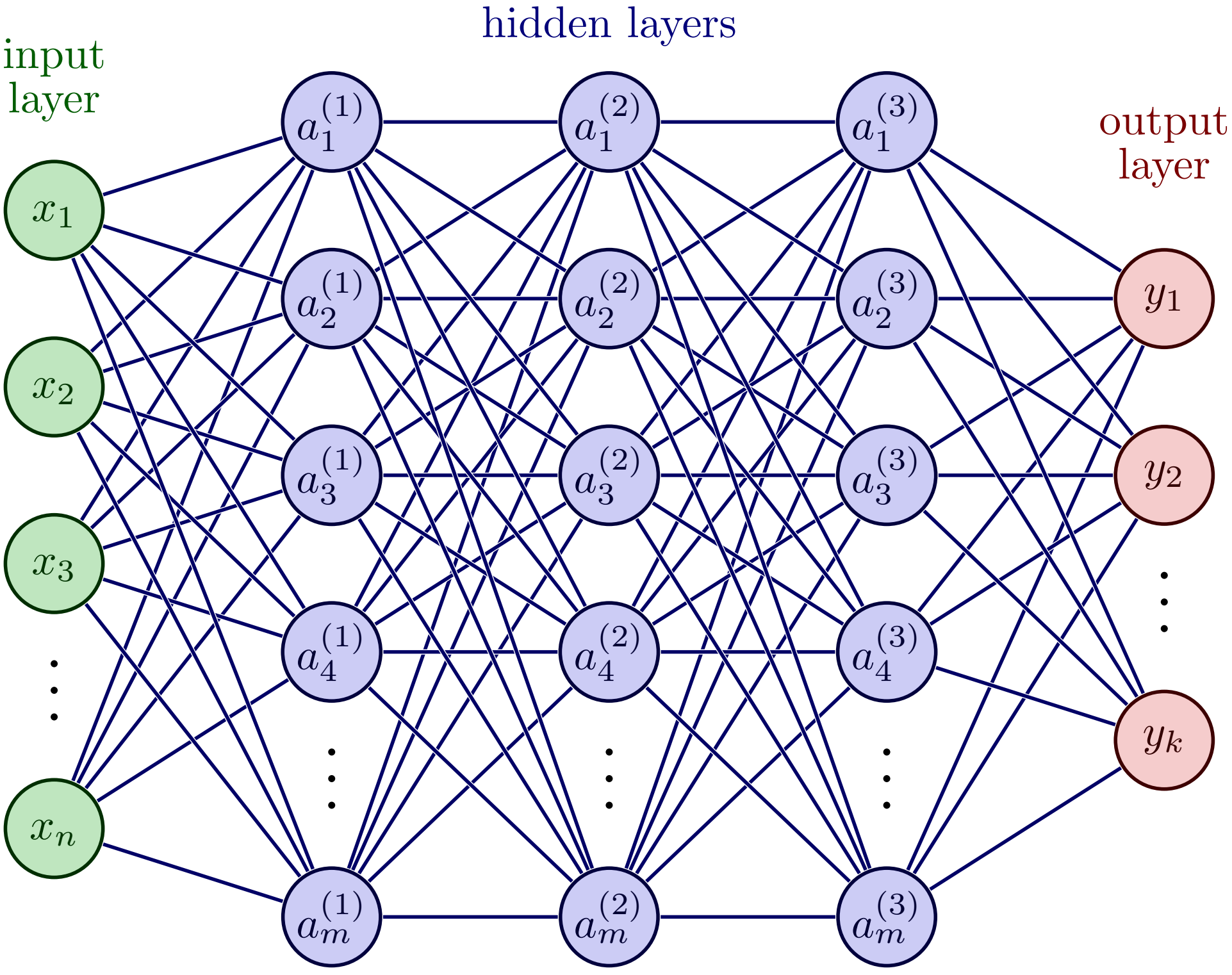
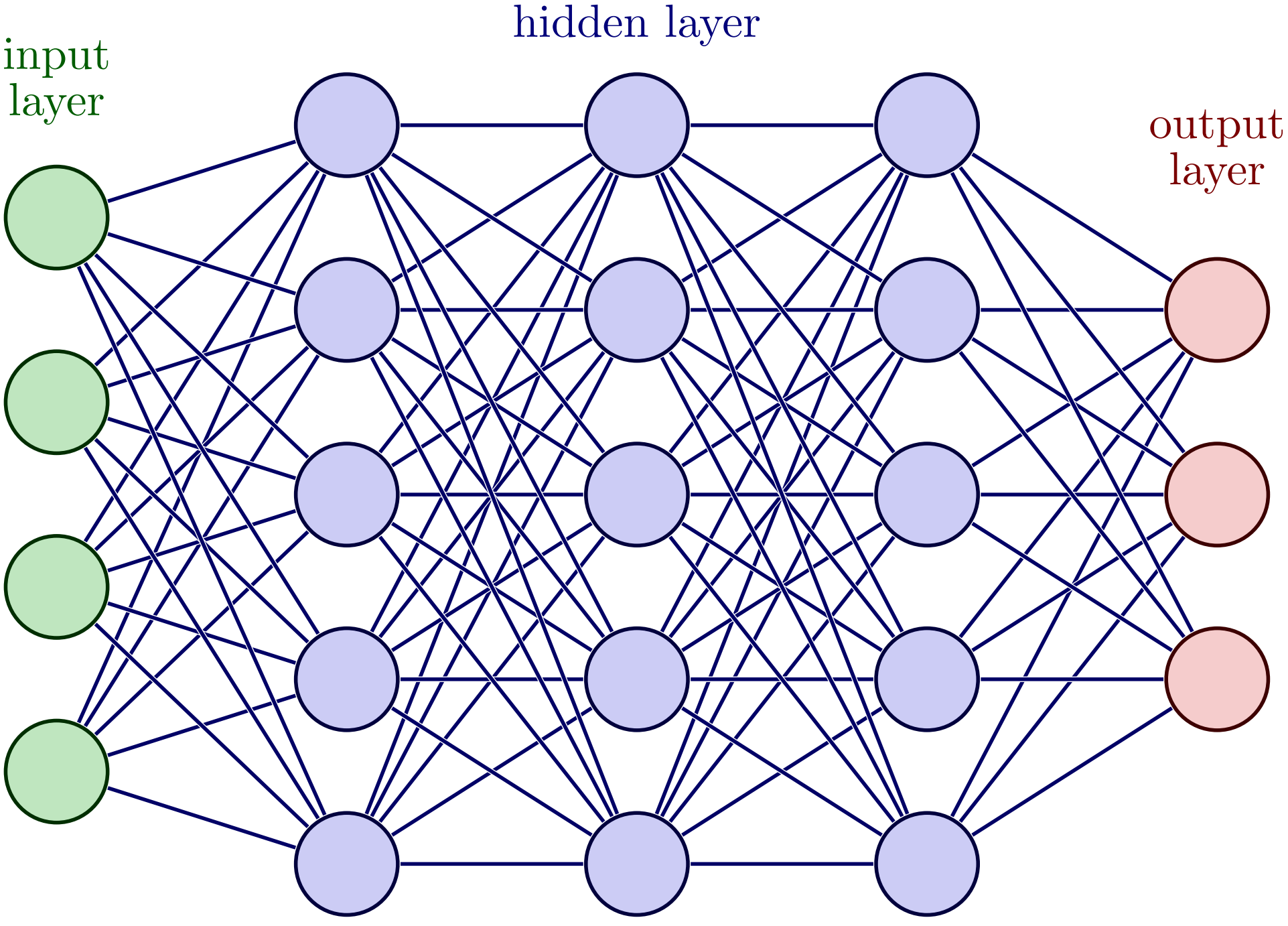
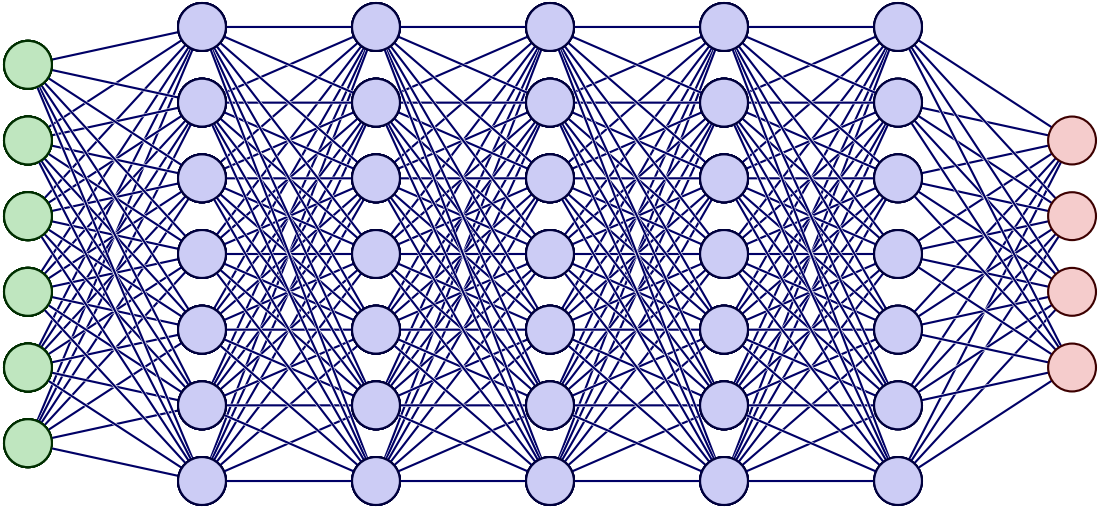

hidden layers
input layer
output layer
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$a_2^{(0)}$
$a_3^{(0)}$
$a_4^{(0)}$
$a_1^{(1)}$
$a_2^{(1)}$
$a_3^{(1)}$
$a_4^{(1)}$
$a_5^{(1)}$
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$a_2^{(2)}$
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$a_3^{(4)}$

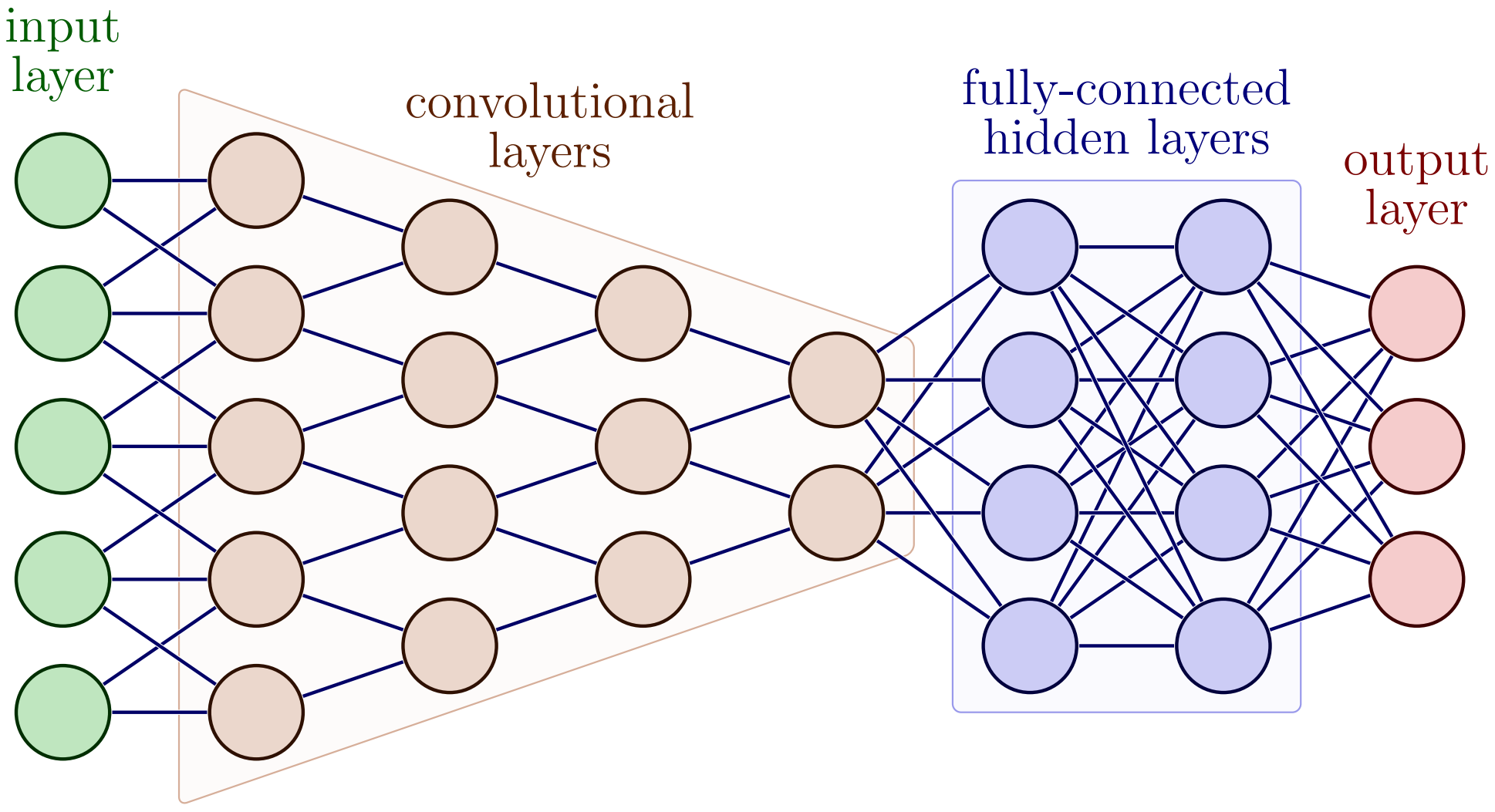

hidden layers
input layer
output layer
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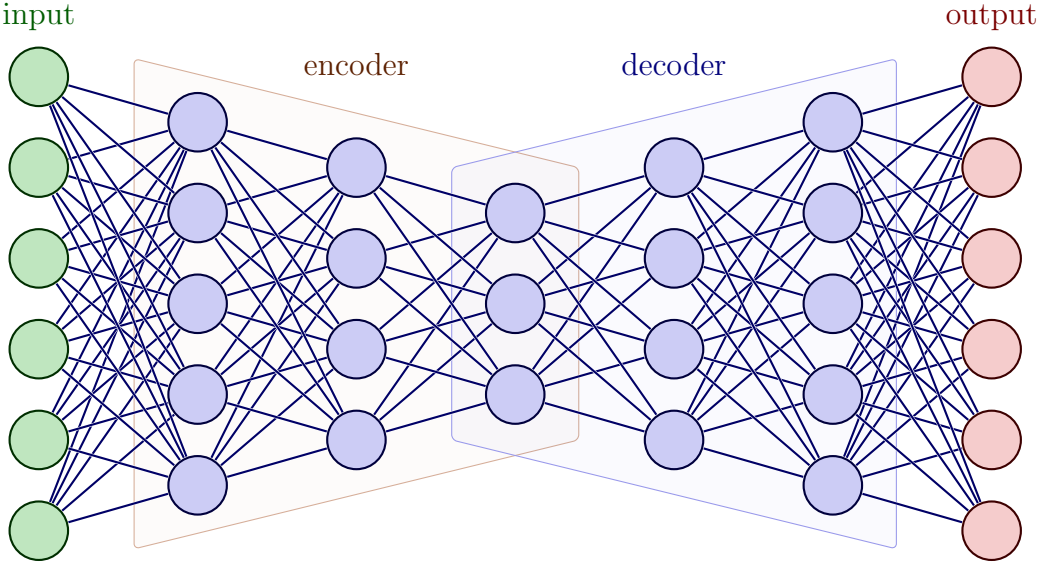

input layer
hidden layer
output layer
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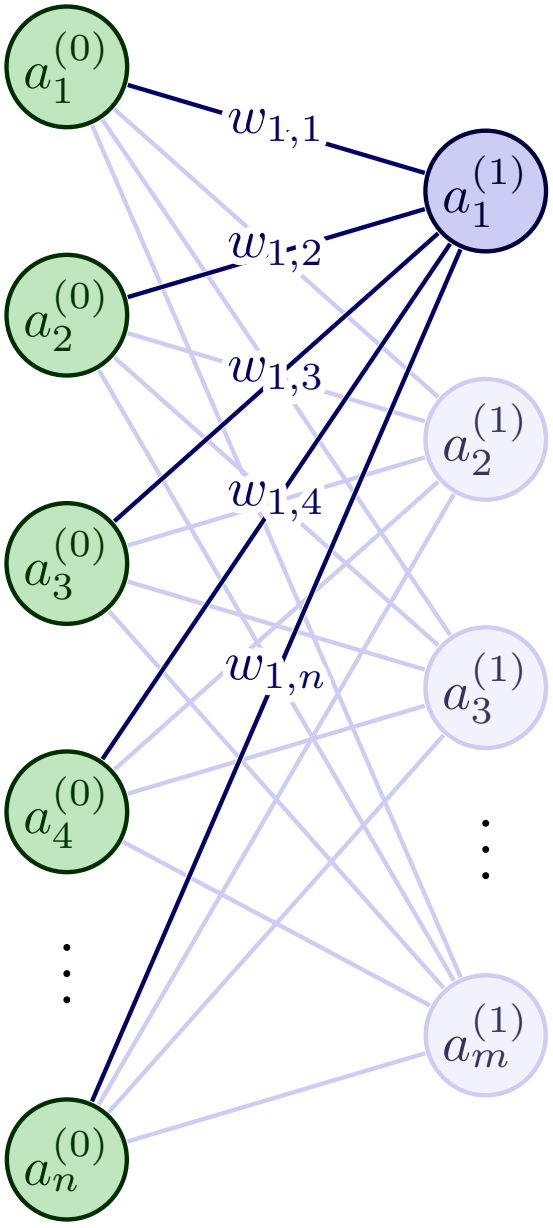

hidden layers
input layer
output layer
x_1
x_2
x_3
x_n
$a_1^{(1)}$
$a_2^{(1)}$
$a_3^{(1)}$
$a_4^{(1)}$
$a_m^{(1)}$
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$a_4^{(2)}$
$a_m^{(2)}$
$a_1^{(3)}$
$a_2^{(3)}$
$a_3^{(3)}$
$a_4^{(3)}$
$a_m^{(3)}$
y_1
y_2
y_k


hidden layer
input layer
output layer




input
layer
convolutional
layers
fully-connected
hidden layers
output
layer


input
encoder
decoder
output



$$= \sigma\left(w_{1,1}a_1^{(0)} + w_{1,2}a_2^{(0)} + \ldots + w_{1,n}a_n^{(0)} + b_1^{(0)}\right)$$

$$= \sigma\left(\sum_{i=1}^{n} w_{1,i}a_i^{(0)} + b_1^{(0)}\right)$$

$$\begin{pmatrix} a_1^{(1)} \\ a_2^{(1)} \\ \vdots \\ a_m^{(1)} \end{pmatrix} = \sigma\left[\begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,n} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m,1} & w_{m,2} & \cdots & w_{m,n} \end{pmatrix}\begin{pmatrix} a_1^{(0)} \\ a_2^{(0)} \\ \vdots \\ a_n^{(0)} \end{pmatrix} + \begin{pmatrix} b_1^{(0)} \\ b_2^{(0)} \\ \vdots \\ b_m^{(0)} \end{pmatrix}\right]$$

$$\mathbf{a}^{(1)} = \sigma\left(\mathbf{W}^{(0)}\mathbf{a}^{(0)} + \mathbf{b}^{(0)}\right)$$